

WEEK 1 REPORT

DATA CLEANING AND TRANSFORMATION REPORT

SUPERSTORE SALES DATA IMPORT, POWER QUERY & DATA CLEANING

Introduction

Microsoft Power BI is a powerful data visualization and data analysis tool used to prepare data and create meaningful reports and dashboards. In this task, I used **Power BI Desktop** and **Power Query Editor** to import and prepare the **Superstore Sales Dataset**.

The dataset was examined for correct data types, missing values, duplicate or unique records, and unwanted spaces or characters in text fields. Various transformation techniques such as renaming columns, splitting columns, and formatting text were performed. I also created useful date-based columns such as **Year, Quarter, Month, Day, and Day of Week** from the **Order Date** column. After completing the cleaning and transformation process, the data was validated and loaded into the Power BI model.

Objectives

- To load the Superstore Sales Dataset into Power BI.
- To understand and explore Report, Data, and Model views.
- To examine the dataset using Power Query Editor.
- To verify and modify incorrect data types.
- To identify blank, null, empty, and error values.
- To check the uniqueness of records.
- To clean text values using Trim and Clean functions.
- To rename and transform the required columns.
- To split columns wherever necessary.
- To generate Year, Quarter, Month, Day, and Day of Week from Order Date.
- To prepare the final cleaned dataset for analysis in Power BI.

Tools and Dataset

Tools and Dataset

Tool Used: Microsoft Power BI Desktop

Data Preparation: Power Query Editor

Dataset: Superstore Sales Dataset

File Format: CSV

Power Query Editor was used as the main data preparation environment to inspect, clean, transform, and organize the Superstore Sales data before using it for report creation.

1. Importing the Dataset

Initially, I opened Power BI Desktop and imported the Superstore Sales CSV file. After selecting the dataset, I opened it in Power Query Editor to perform the required data cleaning and transformation activities.

Steps Followed

1. Launched **Power BI Desktop**.
2. Selected the **Home** tab.
3. Clicked on **Get Data**.
4. Selected **Text/CSV** from the available options.
5. Browsed and selected the Superstore Sales Dataset.
6. Clicked **Open**.
7. Verified the displayed data preview.
8. Selected **Transform Data**.
9. The dataset was opened in **Power Query Editor** for further processing.

samplesuperstore.csv

File Origin

65001: Unicode (UTF-8)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

Row ID

Order ID

Order Date

Ship Date

Ship Mode

Customer ID

Customer Name

Segment

Country/Region

City

1

US-2023-103800

1/3/2023

1/7/2023

Standard Class

DP-13000

Darren Powers

Consumer

United States

Houston

2

US-2023-112326

1/4/2023

1/8/2023

Standard Class

PO-19195

Phillina Ober

Home Office

United States

Naperville

3

US-2023-112326

1/4/2023

1/8/2023

Standard Class

PO-19195

Phillina Ober

Home Office

United States

Naperville

4

US-2023-112326

1/4/2023

1/8/2023

Standard Class

PO-19195

Phillina Ober

Home Office

United States

Naperville

5

US-2023-141817

1/5/2023

1/12/2023

Standard Class

MB-18085

Mick Brown

Consumer

United States

Philadelp

6

US-2023-167199

1/6/2023

1/10/2023

Standard Class

ME-17320

Maria Etezadi

Home Office

United States

Henderso

7

US-2023-167199

1/6/2023

1/10/2023

Standard Class

ME-17320

Maria Etezadi

Home Office

United States

Henderso

8

US-2023-106054

1/6/2023

1/7/2023

First Class

JO-15145

Jack O'Briant

Corporate

United States

Athens

9

US-2023-167199

1/6/2023

1/10/2023

Standard Class

ME-17320

Maria Etezadi

Home Office

United States

Henderso

10

US-2023-167199

1/6/2023

1/10/2023

Standard Class

ME-17320

Maria Etezadi

Home Office

United States

Henderso

11

US-2023-167199

1/6/2023

1/10/2023

Standard Class

ME-17320

Maria Etezadi

Home Office

United States

Henderso

12

US-2023-130813

1/6/2023

1/8/2023

Second Class

LS-17230

Lycoris Saunders

Consumer

United States

Los Angel

13

US-2023-167199

1/6/2023

1/10/2023

Standard Class

ME-17320

Maria Etezadi

Home Office

United States

Henderso

14

US-2023-167199

1/6/2023

1/10/2023

Standard Class

ME-17320

Maria Etezadi

Home Office

United States

Henderso

15

US-2023-105417

1/7/2023

1/12/2023

Standard Class

VS-21820

Vivek Sundaresam

Consumer

United States

Huntsville

16

US-2023-105417

1/7/2023

1/12/2023

Standard Class

VS-21820

Vivek Sundaresam

Consumer

United States

Huntsville

17

US-2023-135405

1/9/2023

1/13/2023

Standard Class

MS-17830

Melanie Seite

Consumer

United States

Laredo

18

US-2023-135405

1/9/2023

1/13/2023

Standard Class

MS-17830

Melanie Seite

Consumer

United States

Laredo

19

US-2023-149020

1/10/2023

1/15/2023

Standard Class

AJ-10780

Anthony Jacobs

Corporate

United States

Springfiel

20

US-2023-149020

1/10/2023

1/15/2023

Standard Class

AJ-10780

Anthony Jacobs

Corporate

United States

Springfiel

Extract Table Using Examples

Load

Transform Data

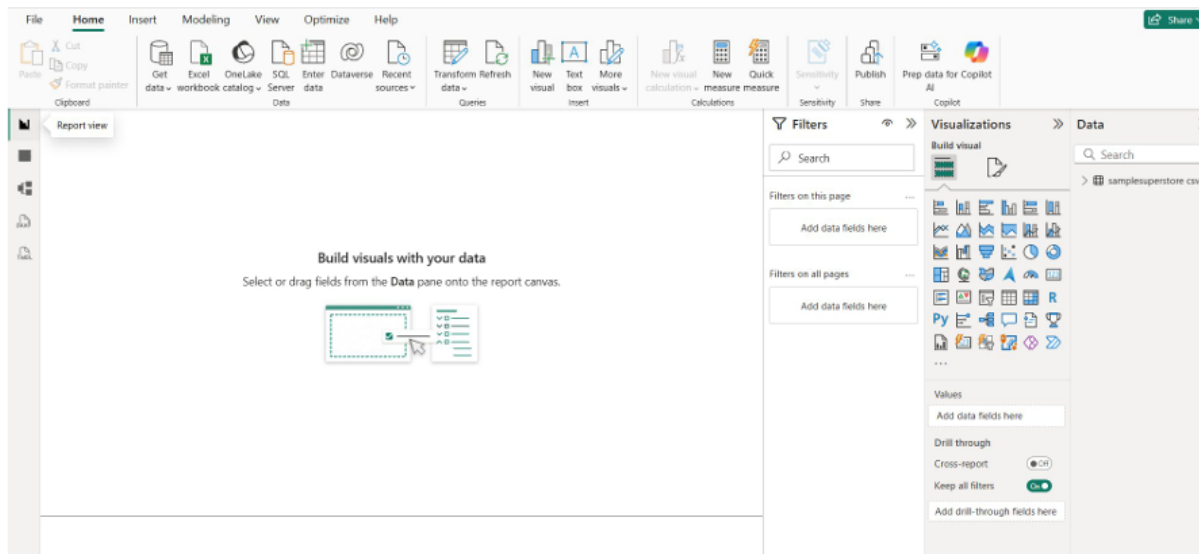
Cancel

2. Exploring Power BI Views

Power BI Desktop provides different views that help in analyzing and managing data.

Report View

Report View is mainly used to design dashboards and reports by adding charts, tables, cards, and other visual elements.



Data View

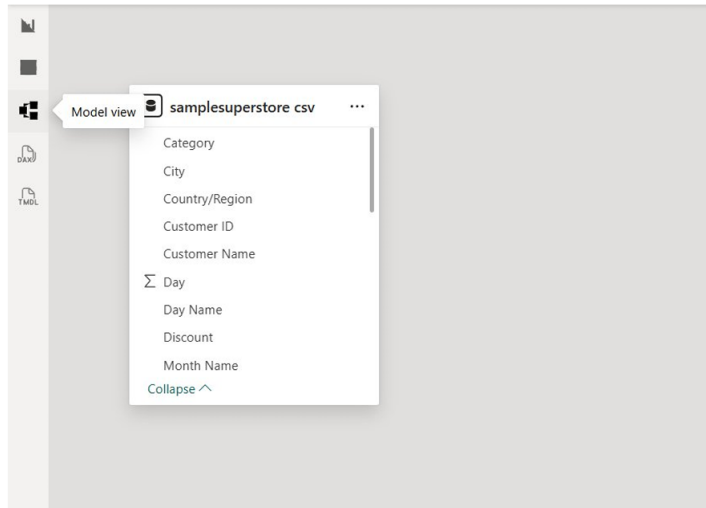
Data View helps to examine the actual records available in the dataset. It allows the user to inspect individual rows and columns.

The screenshot shows the Power BI Desktop interface in Data View. The top ribbon is the same as in the Report View. The main canvas displays a table with columns: Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country/Region, City, State/Province, Postal Code, and Region. The table contains 30 rows of data. On the right side, there is a Data pane with a search bar and a table view icon. The table view icon is selected, and the Data pane shows the same table as the main canvas.

	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State/Province	Postal Code	Region
57	US-2023-148614	20 January 2023	08 January 2023	Standard Class	LS-17230	Lyons Saunders	Consumer	United States	Los Angeles	California	90049	V
122	US-2023-164903	20 February 2023	24 February 2023	Standard Class	SR-20740	Steven Roelle	Home Office	United States	Los Angeles	California	90049	V
331	US-2023-102652	06 April 2023	12 April 2023	Standard Class	AY-10555	Andy Yotov	Corporate	United States	Los Angeles	California	90049	V
332	US-2023-102652	06 April 2023	12 April 2023	Standard Class	AY-10555	Andy Yotov	Corporate	United States	Los Angeles	California	90049	V
370	US-2023-120740	15 April 2023	15 April 2023	Same Day	PS-18970	Paul Stevenson	Home Office	United States	Los Angeles	California	90049	V
373	US-2023-110949	18 April 2023	23 April 2023	Standard Class	IL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	V
375	US-2023-110949	18 April 2023	23 April 2023	Standard Class	IL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	V
377	US-2023-110949	18 April 2023	23 April 2023	Standard Class	IL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	V
723	US-2023-114125	09 July 2023	13 July 2023	Standard Class	GH-14410	Gary Hansen	Home Office	United States	Los Angeles	California	90049	V
727	US-2023-114125	09 July 2023	13 July 2023	Standard Class	GH-14410	Gary Hansen	Home Office	United States	Los Angeles	California	90049	V
1409	US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	V
1410	US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	V
1411	US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	V
1422	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1427	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1428	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1430	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1431	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1446	US-2023-168305	01 November 2023	08 November 2023	Standard Class	PL-18925	Paul Lucas	Home Office	United States	Los Angeles	California	90049	V
1449	US-2023-168305	01 November 2023	08 November 2023	Standard Class	PL-18925	Paul Lucas	Home Office	United States	Los Angeles	California	90049	V
1468	US-2023-153619	03 November 2023	03 November 2023	Same Day	TC-21535	Tracy Collins	Home Office	United States	Los Angeles	California	90049	V
1816	US-2023-138737	07 December 2023	10 December 2023	First Class	FP-14320	Frank Preis	Consumer	United States	Los Angeles	California	90049	V
1859	US-2023-113579	12 December 2023	15 December 2023	Second Class	KD-16345	Katherine Duich	Consumer	United States	Los Angeles	California	90049	V
2288	US-2024-138219	23 March 2024	30 March 2024	Standard Class	BP-11095	Bart Pistole	Corporate	United States	Los Angeles	California	90049	V
3056	US-2024-121188	28 August 2024	04 September 2024	Standard Class	CB-12025	Cassandra Brandow	Consumer	United States	Los Angeles	California	90049	V

Model View

Model View is used to understand the structure of the data model and manage relationships between different tables.



Steps Followed

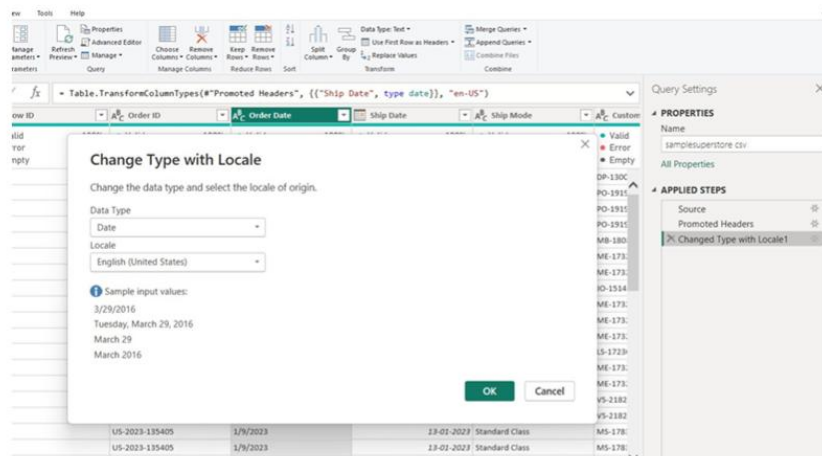
1. Opened the Power BI Desktop application.
2. Viewed the **Report View**.
3. Opened the **Data View** to inspect the dataset.
4. Checked the **Model View**.
5. Reviewed the table structure and available fields.
6. Confirmed that the dataset was loaded correct.

3. Verifying and Modifying Data Types

After importing the dataset, I verified the data type assigned to each important column. Correct data types are necessary to perform accurate calculations and analysis.

Steps Followed

1. Opened the dataset in **Power Query Editor**.
2. Selected the required column.
3. Checked the data type displayed beside the column name.
4. Identified columns that required data type changes.
5. Used **Data Type → Change Type** to modify them.
6. Set categorical fields to **Text**.
7. Set quantity-related fields to **Whole Number**.
8. Set sales and profit fields to **Decimal Number**.
9. Changed the **Order Date** field to **Date**.



4. Cleaning Text Data Using Trim and Clean

Text fields may contain unnecessary spaces or unwanted characters that can affect data analysis. Therefore, I cleaned the categorical columns using the **Trim** and **Clean** options.

The main categorical columns considered were:

- Category
- Sub-Category
- Segment

Steps Followed

1. Selected the **Category** column.
2. Opened the **Transform** tab.
3. Selected **Format**.
4. Applied the **Trim** operation.
5. Checked the values after removing unnecessary spaces.
6. Again selected **Transform** → **Format** → **Clean**.
7. Verified the cleaned values.
8. Applied the same process to **Sub-Category**.
9. Repeated the cleaning process for **Segment**.
10. Reviewed the columns to ensure consistent text formatting.

Trim helps remove unwanted spaces from the beginning and end of text values, while **Clean** removes non-printable characters.

Product ID	Category	Sub-Category	Product Name	Sales
OFF-BJ-10000174	Office Supplies	Binders	Office Supplies	
OFF-BA-10000074	Office Supplies	Binders	Office Supplies	
OFF-BA-10000123	Office Supplies	Binders	Office Supplies	
OFF-ST-10002743	Office Supplies	Storage	Office Supplies	
OFF-AB-10000378	Office Supplies	Art	Office Supplies	
FUR-CH-10004863	Furniture	Chairs	Office Supplies	
OFF-AB-10000662	Office Supplies	Art	Office Supplies	
OFF-AB-10000399	Office Supplies	Art	Office Supplies	
OFF-BJ-10000452	Office Supplies	Binders	Office Supplies	
OFF-FA-10000183	Office Supplies	Fasteners	Office Supplies	
OFF-PA-10000955	Office Supplies	Paper	Office Supplies	
OFF-PA-10000089	Office Supplies	Paper	Office Supplies	
TSC-PR-10000439	Technology	Phones	Office Supplies	
TSC-PR-10000497	Technology	Phones	Office Supplies	
FUR-FU-10000884	Furniture	Furniture	Office Supplies	
OFF-BJ-10000308	Office Supplies	Binders	Office Supplies	
OFF-AB-10000378	Office Supplies	Art	Office Supplies	
TSC-AC-10001186	Technology	Accessories	Office Supplies	
FUR-FU-10000065	Furniture	Furniture	Office Supplies	
OFF-BA-10000172	Office Supplies	Binders	Office Supplies	
FUR-FU-10000030	Furniture	Furniture	Office Supplies	
FUR-BA-10000084	Furniture	Bookcases	Office Supplies	
FUR-CH-10000422	Furniture	Chairs	Office Supplies	
OFF-AB-10000314	Office Supplies	Art	Office Supplies	

5. Checking Blank, Null and Error Values

The dataset was checked for missing or invalid values to ensure that the data was suitable for further analysis.

Steps Followed

1. Selected the required column.
2. Opened the column filter.
3. Examined the values displayed in the filter list.
4. Checked for **blank values**.
5. Checked for **empty values**.
6. Looked for **null values**.
7. Checked whether any **Error** values were present.
8. Reviewed the columns containing unwanted or missing values.
9. Corrected or handled the values wherever necessary.
10. Rechecked the columns after the cleaning process.

This step helped to improve the quality and reliability of the dataset.

Product Name	Text.Trim	type text
Message Book, Wirebound, Four 5 1/2" x 4" Forms/Pg., 200 Dupl. Sets...		
Global Standard Plastic Binding Systems Combs		
Avery 508		
Safco Boltless Steel Shelving		
Avery Hi-Liter Everhold Pen Style Fluorescent Highlighters, 4/Pack		
Global Deluxe High-Back Manager's Chair		
Riggers Handheld Barrel Pen/ Pencil Sharpener		
Dixon Prang Watercolor Pencils, 10-Color Set With Brush		
Stico Hi-Tech Manual Binding System		
Alliance Super-Size Bands, Assorted Sizes		
Southworth 22% Cotton Granite Paper & Envelopes		
Xerox 225		
Wireless Extenders Zboost Yd45 Soho Signal Booster		
Ge 3052464		
Howard Miller 14-1/2" Diameter Chrome Round Wall Clock		
Acico Four Pocket Poly Ring Binder With Label Holder, Smoke, 1"		
Newell 312		
Memorex Micro Travel Drive 8 Gb		
Howard Miller 11-1/2" Diameter Ridgewood Wall Clock		
Avery 482		
Five Value 11-Channel Document Frames, Easel Back		
O'Sullivan Elevations Bookcase, Cherry Finish		
Global Highback Leather Tiller in Burgundy		
4009 Highlighters By Sanford		
Avery Metallic Poly Binders		

6. Performing Column Transformations

Column transformations were carried out to make the dataset easier to understand and use during analysis.

A. Renaming Columns

Steps Followed

1. Selected the required column.
2. Right-clicked on the column.
3. Selected **Rename**.
4. Entered the appropriate column name.
5. Pressed Enter.
6. Verified the updated column name.

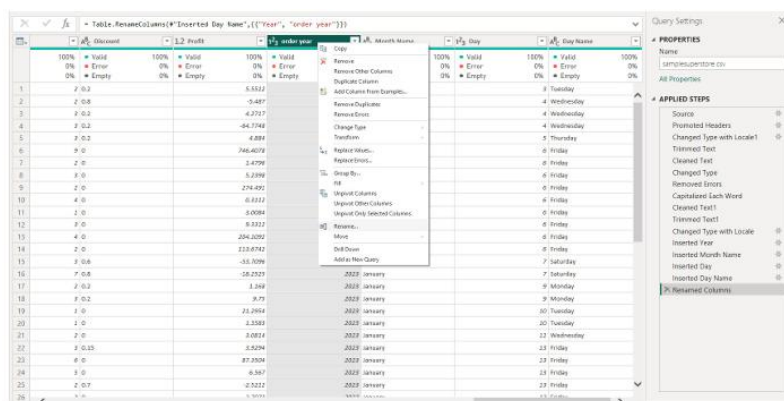
Renaming columns helped to make the dataset fields clear and meaningful.

B. Splitting Columns

Where required, columns containing multiple pieces of information were divided into separate columns.

Steps Followed

1. Selected the required column.
2. Opened the **Transform** tab.
3. Selected **Split Column**.
4. Selected the appropriate splitting option.
5. Defined the required delimiter or position.
6. Applied the transformation.
7. Checked the newly created columns.



7. Creating Date Attributes from Order Date

The **Order Date** column was converted into the **Date** data type. From this column, additional date-related attributes were created to support time-based analysis.

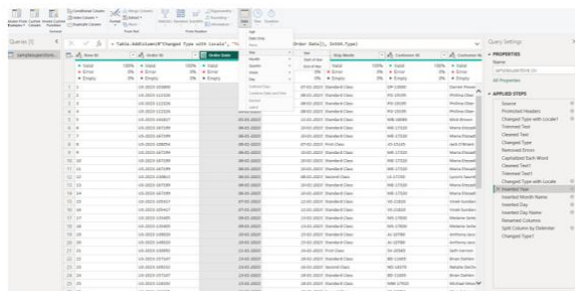
The following attributes were generated:

- Year
- Quarter
- Month
- Day
- Day of Week

A. Creating Year

Steps Followed

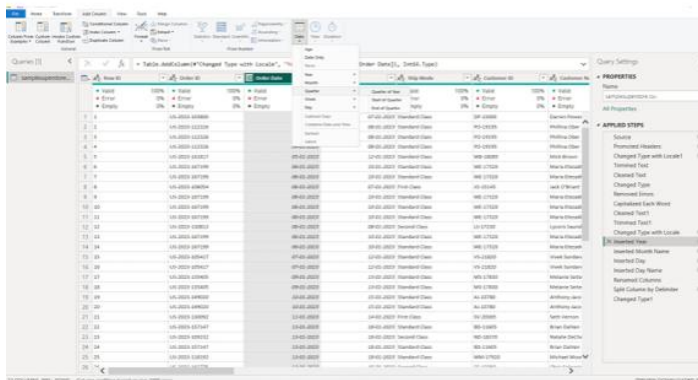
1. Selected the **Order Date** column.
2. Opened the **Add Column** tab.
3. Selected **Date**.
4. Selected **Year**.
5. Chose **Year** from the available option.
6. A new Year column was created.
7. Verified the generated values.



B. Creating Quarter

Steps Followed

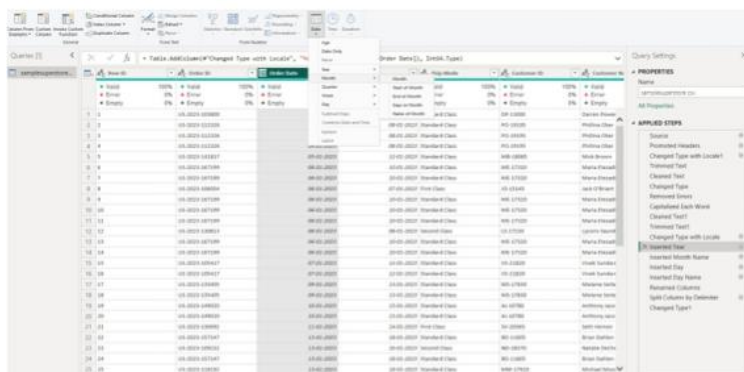
1. Selected **Order Date**.
2. Clicked **Add Column**.
3. Selected **Date**.
4. Selected **Quarter**.
5. Chose **Quarter of Year**.
6. Checked the newly created Quarter column.



C. Creating Month

Steps Followed

1. Selected the **Order Date** field.
2. Went to **Add Column**.
3. Selected **Date**.
4. Selected **Month**.
5. Selected the required month option.
6. Verified the generated Month column.



D. Creating Day

Steps Followed

1. Selected the **Order Date** column.
2. Opened **Add Column**.
3. Selected **Date**.
4. Selected **Day**.
5. Chose the **Day** option.
6. Checked the newly generated Day column.

E. Creating Day of Week

Steps Followed

1. Selected the **Order Date** column.
2. Opened the **Add Column** tab.
3. Selected **Date**.
4. Selected **Day**.
5. Chose **Name of Day**.
6. Verified the Day of Week values.

8. Final Data Validation

After completing all the required cleaning and transformation operations, I performed a final review of the dataset to make sure that the data was accurate and properly prepared for analysis.

Final Checks

- Verified all important data types.
- Confirmed the **Order Date** column was in Date format.
- Verified the **Year, Quarter, Month, Day, and Day of Week** columns.
- Checked for blank and empty values.
- Verified the presence of null values.
- Checked the dataset for any error values.
- Reviewed the cleaned categorical text fields.
- Confirmed that the required columns were renamed correctly.
- Verified the transformed and newly created columns.
- Reviewed the complete dataset before applying the changes.

This final validation ensured that the dataset was clean, consistent, and ready to be loaded into Power BI.

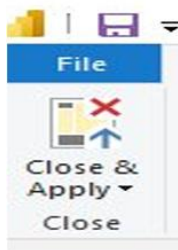
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
1	US-2025-109880	03-01-2025	07-01-2025	Standard Class	OP-13000	Darren Power
2	US-2025-111206	04-01-2025	08-01-2025	Standard Class	PO-13015	Phillina Ober
3	US-2025-111206	04-01-2025	08-01-2025	Standard Class	PO-13015	Phillina Ober
4	US-2025-111206	04-01-2025	08-01-2025	Standard Class	PO-13015	Phillina Ober
5	US-2025-141817	05-01-2025	12-01-2025	Standard Class	MB-18085	Mick Brown
6	US-2025-167199	06-01-2025	10-01-2025	Standard Class	ME-17520	Maria Elscash
7	US-2025-167199	06-01-2025	10-01-2025	Standard Class	ME-17520	Maria Elscash
8	US-2025-109094	06-01-2025	07-01-2025	First Class	JO-15145	Jack O'Brian
9	US-2025-167199	06-01-2025	10-01-2025	Standard Class	ME-17520	Maria Elscash
10	US-2025-167199	06-01-2025	10-01-2025	Standard Class	ME-17520	Maria Elscash
11	US-2025-167199	06-01-2025	10-01-2025	Standard Class	ME-17520	Maria Elscash
12	US-2025-140813	06-01-2025	08-01-2025	Standard Class	LS-17330	Lynne Saund
13	US-2025-167199	06-01-2025	10-01-2025	Standard Class	ME-17520	Maria Elscash
14	US-2025-167199	06-01-2025	10-01-2025	Standard Class	ME-17520	Maria Elscash
15	US-2025-109417	07-01-2025	12-01-2025	Standard Class	VS-21820	Wend Sanders
16	US-2025-109417	07-01-2025	12-01-2025	Standard Class	VS-21820	Wend Sanders
17	US-2025-115405	09-01-2025	13-01-2025	Standard Class	MS-17830	Melanie Sette
18	US-2025-115405	09-01-2025	13-01-2025	Standard Class	MS-17830	Melanie Sette
19	US-2025-149030	10-01-2025	15-01-2025	Standard Class	AJ-10780	Anthony Jacob
20	US-2025-149030	10-01-2025	15-01-2025	Standard Class	AJ-10780	Anthony Jacob
21	US-2025-130902	11-01-2025	14-01-2025	First Class	SV-20365	Seth Versman
22	US-2025-157147	11-01-2025	14-01-2025	Standard Class	BD-11005	Brian Deblen
23	US-2025-109202	11-01-2025	14-01-2025	Standard Class	BD-11005	Brian Deblen
24	US-2025-157147	11-01-2025	14-01-2025	Standard Class	BD-11005	Brian Deblen
25	US-2025-118152	11-01-2025	14-01-2025	Standard Class	MM-17520	Michael Moss

9. Close and Apply

Once all the data cleaning and transformation activities were completed, I applied the changes made in Power Query Editor to the Power BI data model.

Steps Followed

1. Reviewed the complete transformed dataset in **Power Query Editor**.
2. Checked the final changes and transformations.
3. Opened the **Home** tab.
4. Selected **Close & Apply**.
5. Power BI processed and applied all the transformations.
6. The cleaned dataset was successfully loaded into the **Power BI data model**.



Final Result

The **Superstore Sales Dataset** was successfully imported and prepared using **Microsoft Power BI and Power Query Editor**. The required data types were verified and corrected, and text fields were standardized using **Trim** and **Clean** operations.

The dataset was also checked for **blank, empty, null, and error values**. Required column transformations, including **Rename and Split**, were performed. The **Order Date** field was used to generate useful time-based attributes such as **Year, Quarter, Month, Day, and Day of Week**.

After completing all the transformations, the final dataset was validated and loaded successfully into the Power BI data model using **Close & Apply**.

Conclusion

In this task, I successfully performed the basic data preparation process using **Microsoft Power BI and Power Query Editor**. I learned how to import a CSV dataset, inspect its structure, verify data types, and identify missing or incorrect values.

I also learned how to clean text data using **Trim and Clean**, perform column transformations, and create useful date attributes from the **Order Date** column. Finally, I validated the transformed dataset and used **Close & Apply** to load the prepared data into Power BI.

This activity helped me gain practical knowledge of **data cleaning, transformation, and preparation**, which are important steps before creating meaningful reports and dashboards in Power BI.